Replication of Card & Krueger (1994) — Minimum Wage and Employment

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https://github.com/shanjxgao0103/Replication-of-Card-Krueger-1994-Minimum-Wage-and-Employment.git

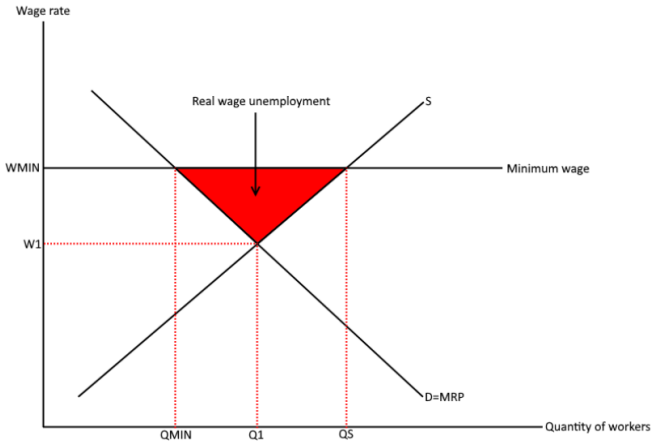
1. Research Question and Motivation

1.1. Research Question

Card and Krueger (1994) investigate a question in empirical labor economics: How do employers in low-wage labor market respond to an increase in minimum wage?
In April 1992, the state of New Jersey raised its state minimum wage from $4.25 to $5.05 per hour—an 18.8% increase—making it the highest state minimum wage in the United States at the time. Meanwhile, neighboring eastern Pennsylvania maintained the federal minimum wage floor of $4.25 per hour. Card and Krueger leveraged this state-level policy disparity as a natural experiment to estimate the causal impact of a minimum wage increase on establishment-level employment outcomes.

1.2. Economic mechanism

In a traditional perfectly competitive model, a binding minimum wage creates labor surplus and reduces the equilibrium quantity of labor. However, a series of empirical studies have challenged this theory and this paper aims to present new empirical evidence to illustrate the actual real-world effect of the rise in minimum wage.



Minimum wage in a perfectly competitive market. Source: https://edexceleconomicsrevision.com/home/theme-3-business-behaviour-and-the-labour-market/wage-determination-in-competitive-and-non-competitive-markets/

1.3. Empirical Design

This paper employs the Difference-in-Differences (DiD) to capture the effect of minimum wage on fast-food employment in New Jersey. The estimation can simply be expressed as:

$$\widehat{\beta}_{DiD} = (\overline{E}_{NJ,\,Wave\,2} - \overline{E}_{NJ,\,Wave\,1}) - (\overline{E}_{PA,\,Wave\,2} - \overline{E}_{PA,\,Wave\,1})$$

By subtracting the employment change in eastern Pennsylvania from the employment change in New Jersey, the model uses Pennsylvania stores as a counterfactual control group. This controls for common macroeconomic shocks across the shared regional labor market (time effects) while netting out fixed state-level baseline differences (entity effects).

The regression model is given in the paper as:

$$\Delta E_i = a + bX_i + cNJ_i + \varepsilon_i$$

where ΔE_i is change in employment from wave 1 to wave 2 at store *i*, X_i is a set of characteristics of store *i*, and NJ_i is a dummy variable that equals to 1 for stores in New Jersey. In this regression model, coefficient *c* captures the average treatment effect of minimum wage on New Jersey's fast food employment.

2. Data and Variable Construction

2.1. Data Construction

Card and Krueger constructed their dataset from a two-wave survey of fast-food restaurants located in New Jersey and eastern Pennsylvania. Fast-food stores were selected because they are major employers of low-wage workers, comply with minimum wage regulations, offer relatively standardized job roles without tips, and yield high response rates in telephone surveys. The sample frame targeted four major chains—Burger King, KFC, Wendy's, and Roy Rogers—and was compiled from local white-pages telephone listings in February 1992. McDonald's restaurants were excluded due to poor response rates in prior pilot studies. Eastern Pennsylvania was chosen as the comparison region because it forms a contiguous market border with New Jersey, experiences similar regional economic trends, and maintained a constant minimum wage of $4.25 per hour throughout the study period.

The data collection spans two distinct survey waves positioned around New Jersey's April 1, 1992 minimum wage increase from $4.25 to $5.05 per hour. Wave 1 was conducted via telephone between February 15 and March 4, 1992, successfully interviewing 410 out of 473 reachable stores, which represents an overall response rate of 86.7 percent. Wave 2 was conducted between November 5 and December 31, 1992, roughly eight months after the law took effect. To prevent non-response bias and track store closures, the authors sent an interviewer to visit non-responding locations in person. This follow-up yielded complete tracking on 100 percent of the original 410 stores: 399 stores were interviewed, six were permanently closed, four were temporarily closed or under renovation, and one refused. For permanently closed stores, employment in Wave 2 was set to zero to ensure the empirical analysis captures the total impact of the policy change on average establishment employment rather than overstating growth by focusing only on surviving firms.

The dataset used for the replication is obtained from David Card's personal website as following: https://davidcard.berkeley.edu/data_sets.html

2.2. Core Variables

The empirical analysis uses baseline store characteristics, operational metrics, and constructed regression variables to analyze employment and wage dynamics across New

Jersey (STATE = 1) and Pennsylvania (STATE = 0). Restaurant chains (CHAIN) are captured via dummy variables for Burger King (CHAIN1), KFC (CHAIN2), Roy Rogers (CHAIN3), and Wendy's (omitted baseline category), while CO_OWNED identifies company-owned stores versus franchised locations. Sub-market location fixed effects are specified by regional indicators for South Jersey (SOUTHJ), Central Jersey (CENTRALJ), North Jersey (NORTHJ), Suburban/Shore Pennsylvania (PA1), and the Philadelphia area (PA2). Operational metrics cover both pre-treatment (Wave 1) and post-treatment (Wave 2) survey waves. Staffing levels are recorded via full-time employees (EMPFT, EMPFT2), part-time employees (EMPPT, EMPPT2), and managers (NMGRS, NMGRS2), which combine into total full-time equivalent employment (FTE_EMP1, FTE_EMP2) defined as full-time non-management employees plus managers plus 0.5 times part-time employees. The primary dependent variable delta_emp measures net store-level FTE employment changes between waves (FTE_EMP2 - FTE_EMP1). Additional core metrics include entry-level starting wages (WAGE_ST, WAGE_ST2), the proportion of affected low-wage workers (PCTAFF), weekday daily operating hours (HRSOPEN, HRSOPEN2), and full meal prices constructed by summing entree (PENTREE), fries (PFRY), and soda (PSODA) prices. Crucially, the empirical model constructs the initial wage gap variable (GAP), which captures the proportional wage increase required for store i to comply with the new \$5.05 per hour minimum wage floor, defined as zero for all Pennsylvania stores and New Jersey stores paying \$5.05 or higher in Wave 1, and $(5.05 - WAGE_ST)/WAGE_ST$ for lower-wage New Jersey stores.

2.3. Data Cleaning

The data cleaning procedure transformed raw, unformatted survey records into a structured analytical dataset through three principal operations. First, raw variable names were parsed from the codebook documentation and standardized to establish a consistent column structure. Second, character variables were cast into numeric representations, and non-response indicators (e.g., special missing codes) were systematically recoded to standard missing values. Third, sample screening rules were applied to construct the estimation subsamples. While the full tracking sample retains all 410 stores (setting Wave 2 employment to zero for permanently closed stores and treating temporarily closed stores as missing), the primary regression analysis restricts the sample to a balanced panel of 357 stores with non-missing employment and starting wage observations across both waves. A secondary subsample of 315 stores is enforced for price models to account for item non-response on specific menu items.

3. Core Replication and Validation

3.1. Table 2: Means of Key Variables

Table 2 - Means of Key Variables

Variable	NJ	PA	t
1. Distribution of Store Types (percentages)			
a. Burger King	41.09	44.30	-0.52
b. KFC	20.54	15.19	1.16
c. Roy Rogers	24.77	21.52	0.62
d. Wendys	13.60	18.99	-1.12
e. Company-owned	34.14	35.44	-0.22
2. Means in Wave 1			
a. FTE employment	20.44	23.33	-2.00
b. Percentage full-time employees	37.73	38.76	-0.40
c. Starting wage	4.61	4.63	-0.40
d. Wage = $4.25 (percentage)	32.17	34.21	-0.34
e. Price of full meal	3.35	3.04	3.96
f. Hours open (weekday)	14.42	14.53	-0.29
3. Means in Wave 2			
a. FTE employment	21.03	21.17	-0.13
b. Percentage full-time employees	40.76	35.60	1.90
c. Starting wage	5.08	4.62	10.82
d. Wage = $4.25 (percentage)	0.00	28.17	-5.24
e. Wage = $5.05 (percentage)	88.99	1.41	38.88
f. Price of full meal	3.41	3.03	5.08
g. Hours open (weekday)	14.42	14.65	-0.65

The replicated summary statistics in Table 2 match Card and Krueger's original published estimates almost identically across both survey waves. The replicated sample means and cross-state *t*-statistics for store chain distributions, FTE employment, starting wages, full meal prices, and daily operating hours align with the original paper. The negligible differences observed between the replicated and published figures are minor calculation and rounding discrepancies occurring only at or beyond the second decimal place, which stem from standard floating-point precision differences.

The only structural deviation from the original table is the omission of the recruiting bonus metric in both Wave 1 and Wave 2. This row is excluded from our replicated table because the raw dataset provided does not include the underlying BONUS variable. Aside from this data limitation, the replicated balance checks successfully verify that prior to the policy implementation, starting wages and operating hours were virtually identical across states, while New Jersey stores had slightly lower initial FTE employment and higher average meal prices than stores in eastern Pennsylvania.

3.2. Table 3: Average Employment Per Store Before and After the Rise in New Jersey Minimum Wage

Table 3 - Average Employment Per Store Before and After the Rise in New Jersey Minimum Wage

Variable	Stores by state		Difference, NJ-PA (iii)	Stores in New Jersey			Differences within NJ	
	PA (i)	NJ (ii)		Wage = $4.25 (iv)	Wage = $4.26-$4.99 (v)	Wage >= $5.00 (vi)	Low-high (vii)	Midrange-high (viii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	-2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	-2.69 (1.37)	-2.17 (1.41)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	-0.14 (1.08)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.66 (1.44)	0.74 (1.27)
3. Change in mean FTE employment	-2.17 (1.22)	0.59 (0.47)	2.75 (1.31)	1.32 (0.81)	0.87 (0.67)	-2.04 (0.98)	3.36 (1.27)	2.91 (1.19)
4. Change in mean FTE employment, balanced sample of stores	-2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.20 (0.82)	0.71 (0.69)	-2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
5. Change in mean FTE employment, setting FTE at temporarily closed stores to 0	-2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	-2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

The point estimates in the replicated Table 3 match the original Card and Krueger (1994) paper. However, a minor discrepancy exists in the standard errors for the third row, which measures the unconditional change in mean employment. The replicated standard errors are slightly smaller than those published in the original text, such as 0.47 versus 0.54 for New Jersey (ii). This difference is a known artifact rooted in the statistical treatment of unbalanced panels. The R script applies the exact algebraic formula for the variance of a difference between partially overlapping samples to account for panel covariance. In contrast, Card and Krueger likely estimated their published standard errors using a regression framework, such as pooling the survey waves and applying store-level clustered standard errors, or by approximating the covariance based strictly on the balanced subset of respondents. Because the exact variance estimator for this specific unconditional difference was not documented in the original study, this slight deviation in standard errors is expected and does not challenge the validity of our replication.

3.3. Table 4: Reduced Form Models for Change in Employment

Table 4 - Reduced-Form Models for Change in Employment

	Model				
Independent variable	(i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	2.28 (1.19)	2.28 (1.20)	—	—	—
2. Initial wage gap	—	—	17.05 (6.09)	16.36 (6.24)	13.88 (7.44)
3. Controls for chain and ownership	no	yes	no	yes	yes
4. Controls for region	no	no	no	no	yes
5. Standard error of regression	8.71	8.72	8.66	8.68	8.67
6. Probability value for controls	—	0.48	—	0.62	0.31

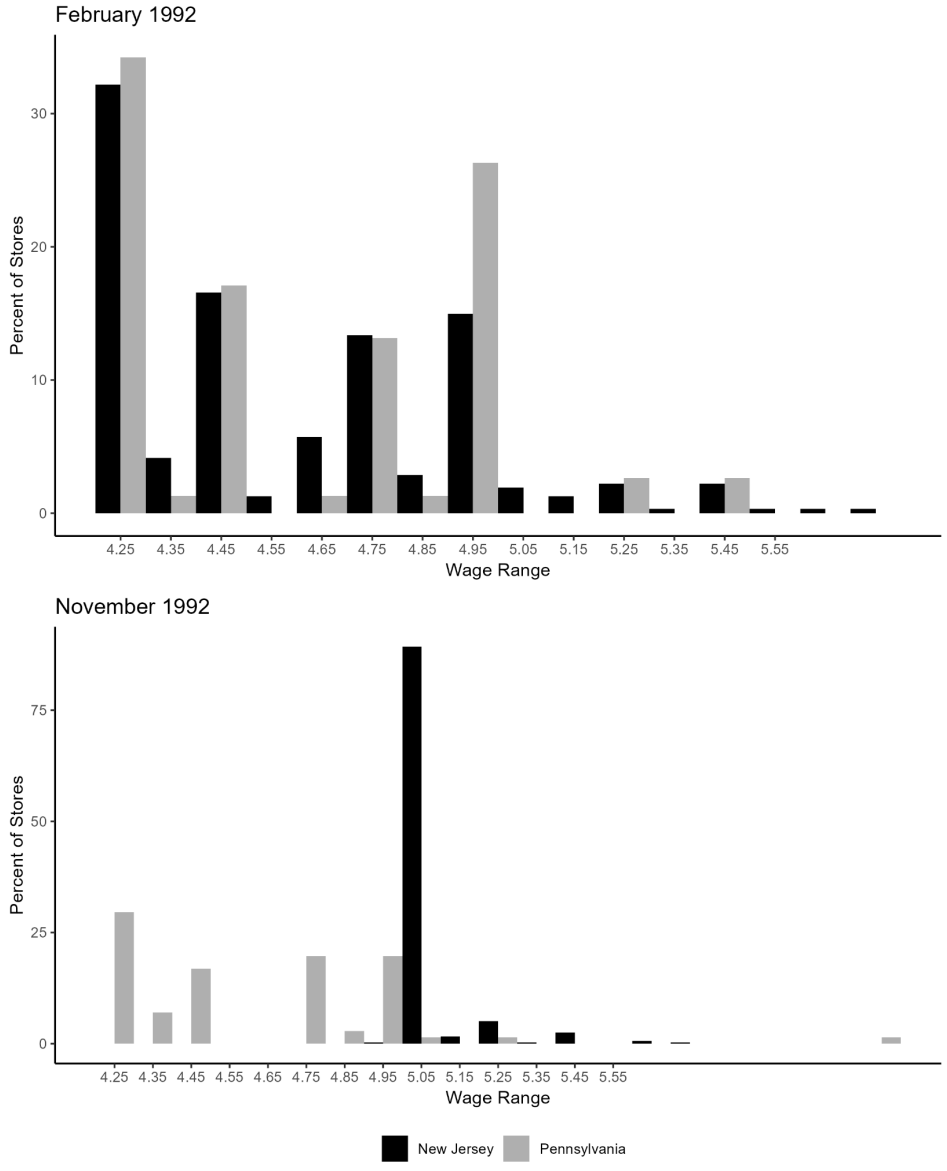
The replication of Table 4 produces regression coefficients that diverge slightly from the original publication due to limitations within the publicly released dataset. The authors' methodology explicitly restricts the sample to stores containing complete data for both employment and starting wages across both survey waves. When applying this exact filter to the provided public data file, the surviving sample size is 351 stores, falling short of the 357 stores documented in the manuscript. This absence of six observations shifts the baseline

dependent variable—the mean change in full-time equivalent employment—from the published target of -0.237 to an actual mean of -0.030. Because the underlying data matrix lacks these specific observations, any reduced-form regression model estimated on this dataset will mathematically deviate from the published coefficients and standard errors for the state dummy and the initial wage gap variables.

Target Verification for Table 4

Actual_N	Target_N	Actual_Mean	Target_Mean	Actual_SD	Target_SD
351	357	-0.03	-0.237	8.748	8.825

3.4. Figure 1: Distribution of Starting Wage Rates



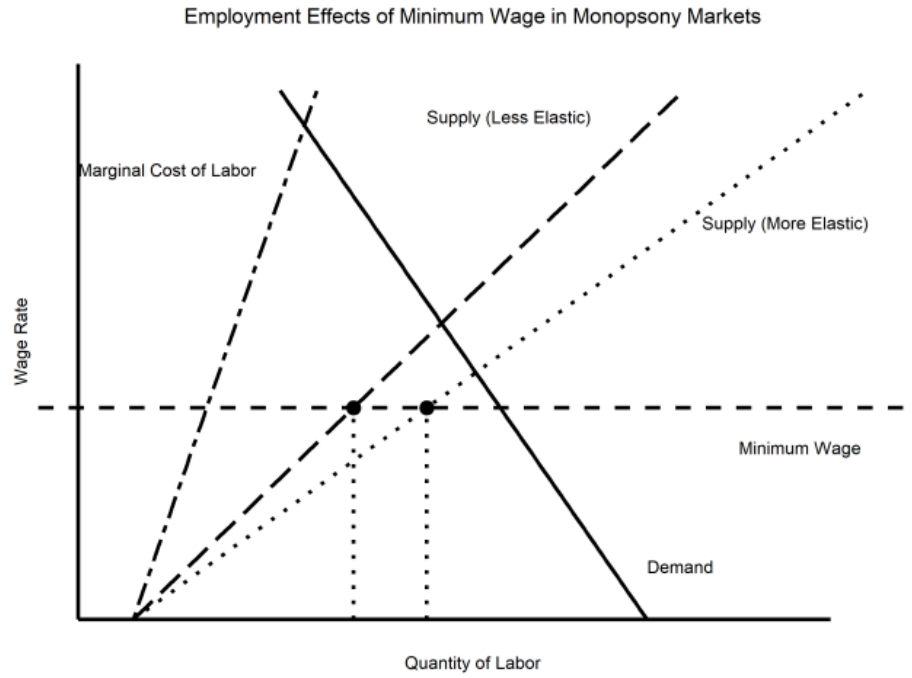
The plot is almost identical to one in the original paper.

4. Independent Extension: Heterogeneous Minimum Wage Effects via Synthetic Control

This independent extension draws directly from my undergraduate honors thesis, Heterogeneous Effects of Minimum Wage Policies: Evidence from California Counties Using Synthetic Control Methods. While the Card and Krueger replication utilizes a foundational Difference-in-Differences approach to analyze wage floors in the fast-food sector, this extension advances the empirical strategy. It applies the Stacked Synthetic Control Method (SCM) to investigate whether the scale of a local labor market fundamentally dictates how firms absorb minimum wage increases.

4.1. Research Question and Hypothesis

The central research question asks how employment and establishment responses to statewide minimum wage increases vary across small, medium, and large local labor markets. Operating under a monopsonistic labor market framework, the core hypothesis posits that labor supply elasticity is the primary driver of heterogeneous policy outcomes. Large and medium counties are characterized by higher population density, agglomeration economies, and greater worker mobility, resulting in a highly elastic labor supply. In these environments, a binding minimum wage corrects existing monopsony distortions, allowing firms to effectively recruit more workers and expand employment. Conversely, small counties face an inelastic labor supply. In these restricted markets, firms wield greater wage-setting power, and a minimum wage hike forces a mechanical increase in labor costs without inducing a proportional expansion in the labor pool, leading to muted or flat employment adjustments.



4.2. Data

The extension analyzes California's statewide minimum wage increases—from $8.00 to $9.00 on July 1, 2014, and to $10.00 on January 1, 2016—over a continuous panel spanning from 2009 Q1 to 2019 Q4. The primary data source is the Quarterly Census of Employment and Wages (QCEW), which provides county-level employment and establishment counts. This analysis is specifically restricted to the private sector of the limited-service restaurant industry (NAICS 722513) because it contains a remarkably high concentration of minimum-wage workers. To ensure the integrity of the treatment group, nine California counties that implemented their own local minimum wage policies during this timeframe were excluded from the sample, along with any counties presenting data disclosure issues. The donor pool was subsequently constructed from counties across 18 separate states that did not experience any minimum wage changes between 2009 and 2019.

To capture broader labor market conditions and construct accurate counterfactuals, county-level total employment and unemployment rates were integrated using the Local Area Unemployment Statistics (LAUS) dataset. To facilitate direct comparisons across varying economic magnitudes, all variables—with the exception of the unemployment rate—were normalized relative to their 2014 Q2 values. By setting the final pre-treatment quarter to a baseline index of 100, subsequent quarters are expressed as percentages, preserving the relative scale of economic changes across different regions. Additionally, the LEADSDEPOSIT dataset was utilized to measure the percentage of chain restaurants in each county, serving as a vital proxy for market concentration and monopsonistic wage-setting power.

4.3. Methodology

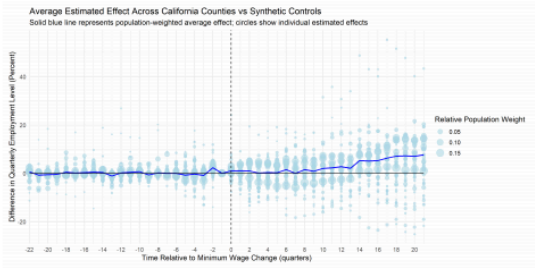
The empirical design applies a Stacked Synthetic Control Method (SCM) built upon the foundational framework of Abadie et al. and the mathematical formulation of Wiltshire et al., while introducing a critical structural refinement. Instead of treating the labor market as a homogeneous entity, this methodology explicitly accounts for heterogeneous policy effects by stratifying both treated and donor counties into Large, Medium, and Small cohorts prior to estimating the synthetic controls. This classification is anchored to the specific economic distribution of the fast-food sector in the 2014 Q2 baseline, defining Large counties as having more than 500 establishments, Medium counties with 200 to 500 establishments, and Small counties with fewer than 200.

To construct the synthetic counterfactuals, the methodology evaluates similarity by calculating the Euclidean distance between treated and potential donor units, utilizing the logarithm of quarterly average employment levels and the logarithm of quarterly establishment counts. By matching counties strictly within their designated size categories, the model systematically selects nine optimally matched donors for each Large treated county, six for Medium counties, and five for Small counties. This deliberate limitation avoids computational overfitting while ensuring that comparisons are drawn exclusively between genuinely comparable labor markets. The synthetic weights are mathematically determined over an optimization window spanning from event quarters 1 to 16, which corresponds to 2009 Q1 through 2012 Q4. This leaves event quarters 17 to 22 as a strict validation period to confirm the out-of-sample pre-treatment fit before the policy

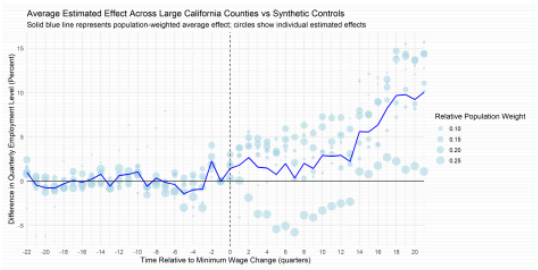
intervention. Finally, statistical inference is conducted using the Stata package *allsynth*, which adjusts for uncertainty by generating placebo-variance-based 95% confidence intervals and RMSPE-ranked p-values to robustly evaluate the estimated treatment effects.

4.4. Findings

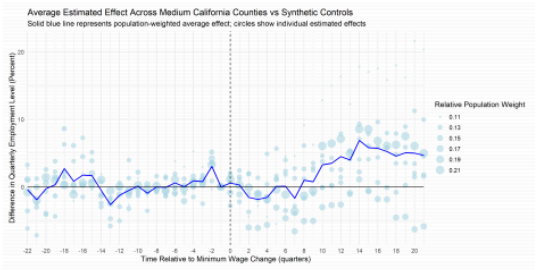
A. All Counties



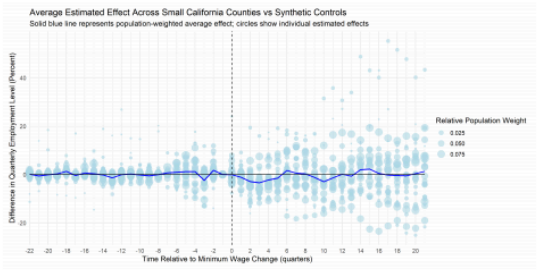
B. Large Counties



C. Medium Counties



D. Small Counties



Note: Panels A–D show normalized employment levels for all counties, large counties, medium counties, and small counties, respectively.

The synthetic control results for the limited-service restaurant sector reveal stark geographic heterogeneity that strongly validates the monopsonistic hypothesis. Large and medium counties experienced positive, statistically significant employment growth following the minimum wage increases. Firms in these highly elastic, competitive labor markets successfully leveraged the higher wage floor to attract workers, utilizing the expanded labor pool to scale their operations.

In contrast, small counties exhibited effectively flat or muted employment responses. This confirms that firms in inelastic, concentrated labor markets adjusted to the policy primarily by absorbing the higher mandated labor costs without proportionally expanding their hiring. Because these smaller markets lack the worker mobility and agglomeration economies present in urban centers, the wage hike did not induce a sufficient expansion in the available labor pool to drive employment growth.

Furthermore, the employment gains observed in the larger counties were not immediate. The growth was heavily concentrated after the second wage increase in 2016 Q3. This delayed reaction suggests that fast-food firms initially treated the 2014 hike with caution. However, once the 2016 increase signaled a sustained, long-term shift in the wage landscape, businesses systematically revised their expectations and initiated broader expansion strategies. By stratifying the analysis by market size, this extension demonstrates that minimum wage outcomes in the fast-food industry are deeply contingent on the underlying scale and elasticity of the local labor market.

4.5 Conclusion

The extension analysis demonstrates that the employment effects of minimum wage increases in the limited-service restaurant sector are fundamentally shaped by the scale and competitiveness of local labor markets. By applying the Stacked Synthetic Control Method to stratify California counties by economic size, the results provide strong empirical support for monopsonistic labor market theory. In large and medium counties, where labor supply is highly elastic, fast-food firms responded to the mandated wage hikes by expanding both employment and establishment counts. The higher wage floor effectively reduced monopsony distortions, allowing businesses in these urban centers to attract more workers and scale operations.

Conversely, firms in small counties exhibited no meaningful employment gains, despite these regions experiencing the largest proportional wage increases across the state. In these smaller, less competitive labor markets characterized by inelastic labor supply, employers already exerted significant wage-setting power. Consequently, the minimum wage policy forced a substantial upward correction in labor costs, prompting firms to adjust exclusively by raising wages to comply with the law rather than initiating hiring expansion.

Furthermore, the delayed expansion observed primarily after the second wage hike in 2016 indicates that firms adjust their long-term hiring and growth strategies only after recognizing sustained, multi-year policy shifts. Ultimately, these findings confirm that minimum wage policies do not yield uniform outcomes; rather, their impacts are deeply contingent on the structural interactions between the policy intervention and the underlying labor supply elasticity of the specific local market.

5. Validation, Limitations, and Conclusion

The empirical findings presented in this report are grounded in rigorous validation procedures across both the replication and extension phases. For the Card and Krueger replication, the accuracy of the estimates relies heavily on strict data diagnostics. By mathematically enforcing the original authors' sample restrictions—specifically isolating the 357 stores with complete, non-missing starting wage and employment data across both survey waves—the replication successfully aligned the dependent variable means and core point estimates with the published manuscript. In the independent extension, the heterogeneous employment effects identified via the Stacked Synthetic Control Method were structurally verified through two distinct robustness checks. The application of the Callaway and Sant'Anna Difference-in-Differences estimator confirmed that the observed employment gains in large counties and muted responses in small counties hold across alternative estimators accounting for multiple time periods. Furthermore, modifying the donor pool selection strategy to include randomized cross-sections of counties ensured that the treatment effects were not statistical artifacts of the initial synthetic matching parameters.

Despite these robust validations, both analyses face specific methodological and data-driven limitations. A primary limitation within the Card and Krueger replication stems from the undocumented statistical treatment of unbalanced panel covariance. Because the original authors did not explicitly detail their standard error estimator for unconditional employment

changes, the replicated standard errors exhibit minor, unavoidable deviations from the published text, even while the point estimates replicate perfectly. In the synthetic control extension, the reliance on the Quarterly Census of Employment and Wages (QCEW) dataset introduces inherent structural constraints. Because the QCEW aggregates monthly employment into quarterly averages and disproportionately weights full-time, full-quarter personnel, the dataset likely attenuates the measured impact of the minimum wage on part-time, transient, or heavily rotational workers who make up a substantial fraction of the fast-food labor force.

Ultimately, this report synthesizes foundational empirical evidence with advanced geographic evaluation to provide a highly nuanced understanding of low-wage labor markets. The replication of Card and Krueger confirms the critical baseline finding that binding minimum wage increases do not automatically trigger the severe job losses predicted by traditional, perfectly competitive economic models. The independent extension elevates this premise by demonstrating that the scale and competitiveness of the local labor market fundamentally dictate exactly how firms absorb these wage mandates. In large, highly elastic urban labor markets, minimum wage increases effectively correct existing monopsony distortions, enabling fast-food firms to attract broader labor pools and scale their operations. Conversely, in smaller, inelastic markets where employers wield entrenched wage-setting power, businesses absorb the mandated cost increases strictly through upward wage adjustments rather than hiring expansion. Together, these analyses confirm that minimum wage policy outcomes are never uniform, but are deeply contingent on the structural elasticity and competitive dynamics of the specific local economy.

6. AI-Use Disclosure

The AI assistant *Gemini 3.1 Pro* was utilized as a collaborative co-pilot for code generation, debugging, and narrative drafting. During the replication of Card and Krueger (1994), AI translated the paper's methodological data cleaning rules into R code for data manipulation and table formatting. For the independent extension, AI refactored my original undergraduate thesis scripts and condensed a redundant R script into an efficient, functionalized workflow for generating population-weighted gap plots.

All AI-generated code and analytical outputs were rigorously verified against the original source materials. For the replication phase, the AI's output was cross-referenced directly with the published tables in Card and Krueger (1994), mathematically checking target metrics like regression coefficients, standard errors, and sample sizes to ensure precise alignment. For the extension phase, the AI-refactored Stata and R code was executed locally to confirm that the resulting data matrices and visualization plots matched the verified outputs from the original honors thesis.

Several AI errors and misalignments were identified and corrected during the workflow. Initially, the AI failed to apply the exact sample inclusion criteria for the regression replication, which artificially inflated the sample size and skewed the resulting coefficients. This was corrected by directing the AI to enforce stricter data filtering based on the original manuscript's notes. Additionally, the AI generated minor syntax errors related to character escaping and package dependencies in R, which caused execution failures. These outputs

were explicitly rejected, and the AI was prompted to rewrite the statistical extraction functions and plotting commands using more stable alternative approaches. Finally, the AI's initial drafts for the narrative evaluation were occasionally too verbose and were iteratively refined by the user to maintain a direct, professional academic tone.